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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION, APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMBG100 - PERSPECTIVES OF MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 10 questions. Each question carries 2 marks

1. How do Streptococci differ from Staphylococci in their arrangement? [U] / [CO1]

Explain arrangement – Streptococci – cocci in chain, Staphylococci – cocci in grape like clusters / Diagram(2)

2. Explain naked and enveloped viruses. [U] / [CO1]

Naked viruses – capsid not enclosed by lipid envelope (derived from host cell), Enveloped viruses possess lipid envelope around the capsid / Diagram (2)

3. Name two common asexual reproductive structures in fungi. [K] / [CO1]

Asexual reproduction occurs through the formation of spores, which can be produced in specialized structures called sporangia (sporangiospores)(1) or conidiophores (conidia)(1)

4. What is endospore? [K] / [CO1]

Some bacteria (Bacillus, Clostridium) form highly resistant resting /dormant state inside the cell to tide over unfavourable conditions like starvation, high temperature, etc. (2)

5. List out the basic components in a bacteriological culture media [K] / [CO2]

Nutrients - Peptone, Meat/yeast extract, Inorganic salts, Water, Gelling or solidifying agent for solid media.(2)

6. Name three special media. [K] / [CO2]

Name 3 special media with one example (Enriched media – Blood agar, Enrichment media – Selenite F broth, Differential media – MacConkey agar)(2)

7. Name any two physical factors affecting growth of bacteria [K] / [CO2]

Temperature (1) & Pressure (1)

8. Classify bacteria based on electron sources. [U] / [CO2]

Define – Lithotrophs (1) & Organotrophs (1)

9. Define microaerophiles. Give examples. [K] / [CO2]

Microaerophiles are damaged by the normal atmospheric level of O2 (20%) and require O2 levels in the range of 2 to 10% for growth (1). Example – Campylobacter/ Lactobacillus/Helicobacter pylori

10. What is the purpose of turbidimetric method ? [K] / [CO3]

Most widely used technique of measuring cell mass – by observing the light scattering capacity of the sample – the amount of light scattered is directly proportional to the biomass of cells present (2)

11. Why does the growth rate decline in the stationary phase? [K] / [CO3]

Growth rate slows as cells enter stationary phase, nutrients become limited and metabolic end products/waste products accumulate. The rate of cell inhibition or death balances out the rate of multiplication The number of cells growing will equal the amount of cells dying (2).

12. Write down the use of the coulter counter. [U] / [CO3]

For accurate cell counting - direct counting of microorganisms, Gives accurate results with larger cells in hospital laboratories – RBC & WBC (2)

[2x10 = 20]

Part B

Short Essay Questions

Answer any 4 questions. Each question carries 5 marks

13. Differentiate between psychrophiles and thermophiles with examples. [U] / [CO2]

Define Psychrophiles, temperature range, natural habitats and examples (2.5). Define thermophiles, temperature range, natural habitats and examples (2.5).

14. What are growth factors, types and importance? Give examples. [K] / [CO2]

Organic growth factors such as vitamins, amino acids, and nucleic acids some growth factors cannot be synthesized by own cellular processes (1). Examples – Thiamine, Riboflavin, Niacin, Biotin, Vitamin K, Pyridoxine, Folic acid, Pantothenic acid, Cobalamin (Any 2 – 1 mark)

Essential growth factors – when growth does not occur in their absence (1.5)

Accessory growth factors – when they enhance growth, not absolutely necessary for growth (1.5).

15. Contrast between Budding and Fragmentation. [U] / [CO3]

Explain Budding, Diagram and example (2.5). Explain Fragmentation, Diagram and example (2.5).

16. Compare and contrast the pour plate, spread plate and membrane filtration methods for bacterial enumeration.- [U] / [CO3]

Pour Plate, Spread Plate, and Membrane Filtration are techniques used to quantify viable bacteria in a sample by calculating colony-forming units (CFU) per mL or gram, but they differ significantly in application and efficiency (1). Sample Application and sample volume -Pour plate -Sample (1 ml) is mixed with molten agar and then poured.Spread plate -Sample (0.1 ml) is spread onto pre-solidified agar. Membrane filtration -Sample is passed through a filter, which is then placed on agar(3).Application (1)

17. Summarize the non living characteristics of viruses. [U] / [CO1]

Explain – Acellular organization, lack the enzymes necessary for protein and nucleic acid synthesis, Lack of metabolism, cannot be grown in artificial media /outside a living host cell. Viruses are inactive, inert and act like non living chemicals, Can be crystallised. Dependence on host machinery (5)

18. Explain budding in yeast. [U] / [CO1]

At the time of budding small portion of cell wall at or near one pole of yeast cell softens and thins, protoplast in this region bulges out in the form of outgrowth or protuberance, gradually increase in size known as bud. As the bud forms nucleus of the parent cell divides with its vacuole. One of the daughter nuclei along with daughter vacuole migrates into the enlarging bud. Bud grows and becomes constricted at the base, cytoplasm of bud and mother cell remains continuous for some time, then the opening between the two cells close, separating the cells leaving a scar on both cells. A scar with convex surface remains on mother cell, daughter cell retains a concave birth scar (3). Diagram (2)

[5x4 = 20]

Part C

Long Essay Question

Answer any 1 questions. Each question carries 10 marks

19. Define total count method and explain the methods used to determine total bacterial count. **[K] / [CO3]**

Explain determination of cell numbers (1). Direct counting methods – Microscopic techniques – Breed & counting chambers (6) Electronic counters (3)

20. What is culture media? Explain different types of culture media based on consistency with examples. [U] / [CO2]

Explain - The percentage of agar used determines the consistency of the medium. Culture media are divided into three types based on consistency; **solid medium, semi-solid medium, and liquid medium – Explain each with example (3 marks each).** Other than these, there are also biphasic media, which consist of both solid and liquid media (1).